

PUNIT MISHRA

Senior ML & AI Tech Lead | Agents, Inference & Multi-Tenant Platforms at Enterprise Scale

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PROFILE

Applied AI engineer and technical leader with 14+ years taking machine learning from prototype to dependable production at enterprise scale. Not only an architect — I **implement, deploy, and own systems end to end**, from agent runtimes and LLM inference pipelines to **GPU deployment on custom H100 clusters**. As Senior ML & AI Tech Lead in SAP CX Data & AI, I *actively lead a team of thirteen* — technically guiding and personally building the agent orchestration, memory, multi-model serving, fine-tuning, and multi-tenant infrastructure beneath our LLM products. Deep command of **Python, Rust**, Kubernetes, distributed inference, vector search, and **secure, ethical AI**.

EXPERIENCE

SAP

Sep 2018 – Present

Senior ML & AI Tech Lead | CX Data & AI

San Ramon, CA

- | **Built and own a domain-agnostic conversational agent platform** — a memory-managed runtime with multi-tenant LangGraph orchestration reusable across business domains, with performance-critical paths written in **Rust**.
- | Authored **natural-language-to-SQL over S/4HANA**, turning business questions into validated queries — now shipping within SAP's **2026 public and private cloud releases**.
- | **Introduced Ray** and built **LLM serving and fine-tuning on custom bare-metal H100 GPU clusters**; multi-model serving (self-hosted plus Claude, Qwen, Granite-class) on Kubernetes with automated evaluation harnesses.
- | Delivered visual and textual LLM agents across SAP's **CX & Commerce AI Toolkit** (PyTorch on Kubernetes); visual-attribute prediction on OpenAI / CLIP embeddings backed by **vector search and RAG**.
- | **Actively lead a team of thirteen** across three workstreams — technically guiding architecture (ADRs), owning AI ethics & compliance, and setting the group's SDLC and engineering standards. **SAP Security Expert**.
- | Designed **multi-tenant ML** on KubeFlow & Knative (Azure AKS) across SAP CX cloud (C4C, CPQ, CDC), with feature stores (Feathr, Feast).
- | Re-architected **vision** — attribute extraction, object detection, visual search — into first-class **gRPC agent services** exposed through an agent gateway and MCP-based service hub.
- | Contribute to **open-source GPU / inference tooling at the kernel level**, upstreaming improvements that harden the self-hosted serving stack.
- | Designed multi-tenant identity and provisioning unifying **IAS and XSUAA**; serve as **forward-deployed engineer** on the platform's first production enterprise customer.

CALLIDUS CLOUD

Feb 2014 – Sep 2018

Senior Software Engineer | Thunderbridge AI / SAP Sales Cloud

SF Bay Area (acquired by SAP, \$2.4B)

- | Designed and built the **Thunderbridge AI** backend in Python on **Dataiku**, with multi-tenancy running every customer workload for SAP Sales Cloud optimization.
- | Ported the full Dataiku ML stack to **GKE on-prem (Google Anthos)** with Istio, and built the CI/CD backbone — Jenkins on Kubernetes, Kaniko image builds, and Artifactory distribution.
- | **Built developer-productivity tooling** — automated pull-request merges with Palantir Bulldozer and Policy-Bot on Kubernetes, wired into SAP GitHub Enterprise via OAuth to capture all Git events — cutting manual merge overhead across the team.
- | Engineered multi-partition data resiliency for ML across **Oracle, Vertica, SAP HANA, and HDFS** (Spark); shipped inference services for CPQ on Dataiku's API node.

- | Integrated SAP security scans (**Coverity**, **Black Duck**, **Fortify**) into Jenkins per commit. As Senior Automation Engineer (earlier), built TDD / Scala ETL validation and resilient Selenium/Java and Node.js UI test frameworks over a self-maintaining Selenium Grid.

IBM

Automation Systems Engineer

Oct 2012 – Dec 2013
Dallas–Fort Worth, TX

- | Built a reporting and **data-validation automation framework** for JPMorgan Chase.
- | Automated VM creation/configuration via a WCF SOAP service (IIS); parallelized test execution and scheduled runs across cloud machines with varied OS/browser configs.
- | Developed reusable **Selenium** / **Java** test APIs (JUnit, TestNG) adopted across the firm.

UC BERKELEY LAW IST

Web Developer

Sep – Dec 2011
Berkeley, CA

- | Built an online teaching-evaluation system for the Law School on a PHP / Oracle 10g backend, integrated with **CalNet** (Berkeley authentication) for end-of-course student evaluations.

LAWPIVOT INC. (GOOGLE STARTUP LABS)

Software Engineering Intern

May – Aug 2011
Mountain View, CA

- | Built user-to-user interaction features on the **Django** backend and an internal analytics tool visualizing user data (logins, time on site) used by the sales team.

LAWPIVOT INC.

Software Engineering Intern

Oct 2009 – Aug 2010
Fremont, CA

- | Automated testing with Selenium IDE and tracked defects; worked with the founding team on wireframes and AWS server configuration.

SELECTED PROJECTS & OPEN SOURCE

- | **CoViber** — local-first memory for LLMs over MCP: a continuously-updated work graph, urgency-triage queue, and local semantic memory, entirely on-device (no cloud egress). *Python*.
- | **Vaultic** — hardware-backed (FIDO2 / YubiKey) password manager in **Rust**; one on-disk vault format across CLI, TUI, daemon, GUI, and an MCP server. *Homebrew tap*.
- | **Shield AI** — AI-powered DNS filtering with real-time threat / intrusion detection.
- | **Pi Race** — zero-dependency **Rust** CLI benchmarking four classic π algorithms (Ramanujan, Euler/Basel, Wallis, Leibniz) on accuracy vs. speed.

SKILLS

AI & AGENTS	Agent orchestration (LangGraph, LangChain), MCP (agent gateway / tool hub), LLM serving, fine-tuning & evaluation, NL-to-SQL, RAG, retriever-reranker, PyTorch, OpenAI / CLIP, MLOps (Kubeflow, Feathr, Feast)
GPU & SERVING	Ray (distributed), custom bare-metal H100 clusters, self-hosted LLM / embedding serving, Knative / KServe
INFRA & CLOUD	Kubernetes, Kyma / Gardener, Istio, Argo / GitOps, Jenkins, Docker, CI/CD; vector DBs (Qdrant, FAISS), Redis; Azure (AKS), GCP (GKE / Anthos)
PLATFORM & DATA	gRPC / protobuf, CAP / XSUA (OIDC / PKCE); PostgreSQL, SAP HANA, Oracle, Vertica, HDFS / Spark, Kafka
LANGUAGES	Python (FastAPI, Django), Rust , Java, Scala, TypeScript / Next.js
DEV PRODUCTIVITY	GitHub Enterprise automation (Bulldozer, Policy-Bot), Artifactory, Kaniko, ADRs, ephemeral PR environments
SECURITY & ETHICS	SAP Security Expert; internal-auth / identity design, threat modeling, SAST / SCA (Coverity, Black Duck, Fortify); AI ethics, safety & compliance

EDUCATION

UNIVERSITY OF CALIFORNIA, BERKELEY — B.S. Electrical Engineering & Computer Science

2012